

ENTHUSIAST ADVANCE COURSE**LEADER PHASE - MED-I (B)****TARGET - NEET MEDICAL 2024**Test Type : **MINOR**Test Pattern : **NEET (UG)****TEST DATE : 23-04-2023****ANSWER KEY**

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	4	3	4	3	3	2	1	1	1	4	2	1	2	2	2	3	2	1	3	2	2	1	1	3	3	2	4	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	2	2	2	2	1	2	3	1	3	3	2	1	4	3	4	4	4	2	3	2	2	2	4	4	2	1	4	2	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	2	1	3	1	4	2	4	3	2	2	3	3	3	4	1	2	3	4	3	2	1	1	1	2	1	3	4	1	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	1	4	1	4	1	3	1	1	2	3	3	4	1	3	2	3	4	4	2	3	2	4	2	2	3	1	1	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	1	2	4	3	3	3	3	2	2	4	4	2	4	4	4	4	3	4	3	3	4	3	3	3	2	3	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	2	1	3	3	4	3	4	2	4	1	1	1	4	3	3	2	4	2	3	1	3	2	4	3	1	2	4	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	3	1	2	2	2	1	2	1	4	3	4	1	4	2	3	1	2	4	4										

HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**3. **Ans (4)**

Theoretical

4. **Ans (3)**

Charge spherical conductor is equipotential

So potential inside at every point = Potential of surface

$$= \frac{kQ}{R}$$

5. **Ans (4)**

Conducting surface behaves as equipotential surface.

8. **Ans (2)**

Potential decreases in the direction of electric field.

9. **Ans (1)**

$$W = q\Delta V$$

as ΔV is same in all conditions, work will be same.10. **Ans (1)**Torque $\vec{\tau} = \vec{p} \times \vec{E}$ then torque on $\vec{p}_2$ due to $\vec{p}_1 = \vec{p}_2 \times \vec{E}_1$ but $\vec{p}_2 \parallel \vec{E}_1$ so $\vec{\tau}_{21} = \vec{0}$ 13. **Ans (2)**

The charge on the outer surface's of plate A and

$$\text{plate C will be} = \frac{Q + 3Q - 2Q}{2} = Q.$$

So the charge on left surface of plate 'A' = Q

∴ Charge on right surface of 'A' will be zero on left surface of plate 'B' would be zero.

∴ Charge on right surface of 'B' will be 3Q

15. **Ans (2)**

Charge on each face of conducting sheet is $2 \times 10^{-8} \text{ C}$

$$\therefore \sigma = \frac{Q}{A} = \frac{2 \times 10^{-8}}{0.5 \times 0.5}$$

$$\sigma = 8 \times 10^{-8} \text{ C/m}^2$$

16. **Ans (2)**

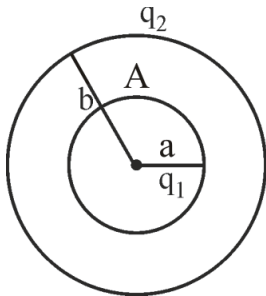
$$\Delta U = q_0 (\Delta V)$$

$$= e (-E.d)$$

$$= (1.6 \times 10^{-19}) (-8 \times 10^4) (0.5)$$

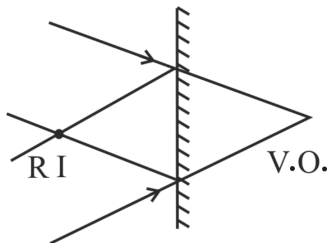
$$= -6.4 \times 10^{-15} \text{ J}$$

17. **Ans (2)**



$$V_a - V_b = q_1 \left[\frac{1}{a} - \frac{1}{b} \right]$$

18. **Ans (3)**

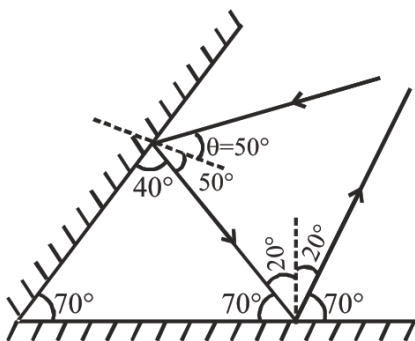


Incident rays are converging type

19. **Ans (2)**

$$\delta = 360 - 2(70^\circ) = 360 - 140 = 220$$

20. **Ans (1)**



21. **Ans (3)**

$$V_I = 2V_m - V_o$$

$$V_o = 0 \quad V_I = 2V_m = 2 \times 10$$

$$= 20 \text{ cm/sec}$$

22. **Ans (2)**

$$\vec{v}_i + \vec{v}_0 = 2\vec{v}_m$$

$$\vec{v}_i = 2\vec{v}_m - \vec{v}_0$$

23. **Ans (2)**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-20} = \frac{1}{20} \Rightarrow v = -10 \text{ cm}$$

24. **Ans (1)**

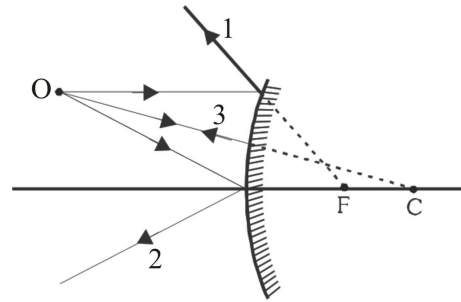
$$2 = \frac{-20}{-20 - u_1} \Rightarrow u_1 = -10 \text{ cm}$$

$$-2 = \frac{-20}{-20 - u_2} \Rightarrow u_2 = -30 \text{ cm}$$

$$\therefore \text{Required distance} = 30 - 10 = 20 \text{ cm}$$

25. **Ans (1)**

Correct Ray diagram is :



26. **Ans (3)**

Concave mirror can form both real and virtual images of real object but it can form only real image of virtual object.

SECTION-B

37. **Ans (2)**

$$V_o = \frac{KQ}{R}$$

$$V_A = \frac{KQ}{6R}$$

$$W_{\text{ext}} = -Q[V_o - V_A]$$

$$= -Q \left[\frac{KQ}{R} - \frac{KQ}{6R} \right]$$

$$= -\frac{5KQ^2}{6R}$$

40. **Ans (3)**

Assertion is correct but Reason is false because under electrostatic condition free electron will be present inside the conductor.

41. Ans (3)

$$U = -\vec{P}_2 \cdot \vec{E}_1 = -P_2 \left(\frac{KP_1}{r^3} \right) \cos 180 = \frac{KP_1 P_2}{r^3}$$

$$F = -\frac{du}{dr} = -\left(KP_1 P_2 \left(\frac{-3}{r^4} \right) \right) = \frac{3KP_1 P_2}{r^4}$$

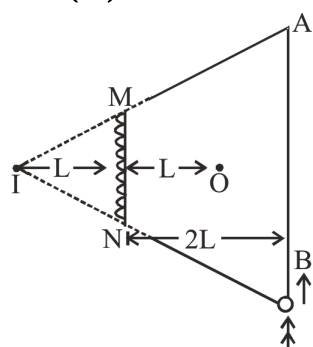
42. Ans (2)

$$V_P = K \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} \right)$$

$$= 9 \times 10^9 \left[\frac{2 \times 10^{-6}}{4} + \frac{(-6 \times 10^{-6})}{5} \right]$$

$$= -6.29 \times 10^3 \text{ V}$$

44. Ans (4)



$$\Delta AIB \approx \Delta MIN$$

$$\frac{MN}{AB} = \frac{L}{3L}$$

$$\frac{d}{AB} = \frac{L}{3L}$$

$$AB = 3d$$

45. Ans (3)

$$v_{Ix} = -2(30 \cos 53^\circ) - 20 \cos 37 = -52 \hat{i}$$

$$v_{Iy} = 20 \sin 37 = 12 \hat{j}$$

46. Ans (4)

Always

50. Ans (3)

$$m = +5 = \frac{f}{f-u} \Rightarrow u = -24 \text{ cm}$$

$$-m = -5 = \frac{f}{f-u} \Rightarrow u = -36 \text{ cm}$$

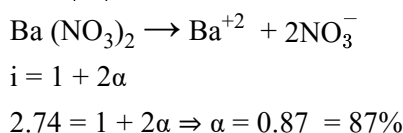
SUBJECT : CHEMISTRY

SECTION-A

51. Ans (2)

$$i = 4$$

52. Ans (2)



53. Ans (2)

$$i = 1 + \alpha(n-1)$$

$$\Rightarrow i = 1 + 0.8(2-1)$$

$$\Rightarrow i = 1.8$$

$$M = \frac{(\%w/v) \times 10}{\text{Mol. wt. of solute}}$$

$$M = \frac{1 \times 10}{74.5}$$

$$\pi = iMRT$$

$$\pi = 1.8 \times \left(\frac{10}{74.5} \right) \times 0.082 \times 300$$

$$\pi = 5.9 \text{ atm}$$

55. Ans (4)

2 mol solute is dissolved in 1 kg water (n = 55.5) therefore, mole fraction of solvent = $\frac{55.5}{2 + 55.5} = 0.965$

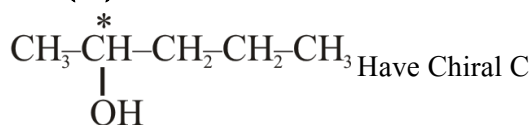
56. Ans (2)

$$\frac{P^0 - P_s}{P_s} = \frac{\text{Moles of solute}}{\text{Moles of solvent}}$$

$$\frac{750 - 740}{740} = \frac{2/M}{156/78}$$

$$M = 74$$

71. Ans (2)



SECTION-B

91. Ans (4)

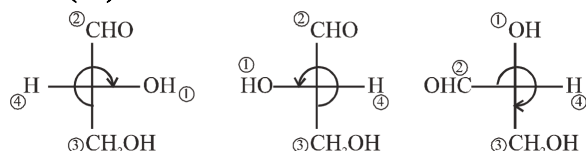
$$V.P. \propto \frac{1}{b.p.}$$

Solution III has lowest Value of V.P. so highest b.p.

92. Ans (1)

Both Assertion & reason are correct & R is correct explanation of the A.

95. **Ans (4)**



- [A] (R) [B] (S) [C] (R)
- (1) A and B are enantiomers
(2) A and C are identical
(3) B and C are enantiomers
(4) A and B are enantiomers

SUBJECT : BIOLOGY-I

SECTION-A

107. **Ans (3)**

Module Pg. # 161

Interferons are made up of 270 amino acid.

110. **Ans (2)**

NCERT Pg No. # 151 (Fig. 8.4)

111. **Ans (3)**

NCERT XII Pg. # 151

112. **Ans (2)**

NCERT (XIIth) Pg. # 151 (para-8.2.2)

113. **Ans (4)**

NCERT (XII) Pg. # 150,151, Para 8.2.1

114. **Ans (2)**

NCERT (XII) Pg. # 151, Para 8.2.2

115. **Ans (2)**

NCERT (XII) Pg. # 153, Para-8.2.7

117. **Ans (1)**

NCERT XII, Pg. # 150

118. **Ans (1)**

NCERT XII, Pg. # 145

119. **Ans (3)**

NCERT XII, Pg. # 146

120. **Ans (3)**

NCERT Pg. # 150 (E)

121. **Ans (4)**

NCERT Pg. # 154 (E)

122. **Ans (3)**

NCERT Pg. # 151 (E)

123. **Ans (1)**

NCERT (XII) Pg. # 151, Para-2

124. **Ans (2)**

NCERT (XII) Pg. # 150,151, para-8.2.1

125. **Ans (4)**

NCERT (XIIth) Pg. # 152 (Para 8.2.3)

126. **Ans (3)**

NCERT XII, Page # 152

127. **Ans (3)**

NCERT XII Pg. # 151 Fig. 8.4

128. **Ans (3)**

NCERT Pg No. # 150 (Para 8.2.1)

129. **Ans (3)**

NCERT XII, Pg. # 146

130. **Ans (2)**

NCERT (XII) Pg. # 154

131. **Ans (2)**

NCERT (XIIth) Pg. # 150-152 (Para 8.2)

SECTION-B

140. **Ans (4)**

NCERT (XII) Pg. # 150, Para-8.2.1

141. **Ans (3)**

NCERT-XII Pg # 151; Para-8.2.2

142. **Ans (3)**

NCERT XII, Pg. # 150

143. **Ans (4)**

NCERT-12th Pg No. # 151

144. **Ans (3)**

NCERT Pg. # 151 (E)

145. **Ans (3)**

NCERT (XIIth) Pg. # 150-152 (Para 8.2)

146. **Ans (3)**

IgG, IgM

NCERT (XII) Pg. # 151, Para-8.2.2

147. **Ans (2)**

NCERT (XII) Pg. # 150, Para-4

SUBJECT : BIOLOGY-II

SECTION-A

157. **Ans (4)**

NCERT Pg. # 244

160. **Ans (2)**

NCERT Pg.# 253-54

163. **Ans (1)**

NCERT Pg. # 220, Fig. 13.1

165. **Ans (4)**

NCERT Pg. # 264

166. **Ans (3)**

NCERT Pg. # 263

167. **Ans (3)**

NCERT Pg. # 264

168. **Ans (2)**

NCERT Pg. # 266

169. **Ans (4)**

NCERT Pg. # 259

170. **Ans (2)**

NCERT Pg. # 266

172. **Ans (1)**

NCERT Pg. # 263

174. **Ans (2)**

NCERT Pg. # 267

175. **Ans (4)**

NCERT Pg. # 263

176. **Ans (3)**

NCERT Pg. # 262

183. **Ans (1)**

NCERT Pg. # 259

184. **Ans (2)**

NCERT Pg. # 260

185. **Ans (2)**

NCERT Pg. # 265

SECTION-B

189. **Ans (1)**

NCERT Pg. # 254

191. **Ans (3)**

Module 3 Pg.# 82

193. **Ans (1)**

NCERT Pg. 221

197. **Ans (1)**

NCERT Pg. # 261

198. **Ans (2)**

NCERT Pg. # 266

200. **Ans (4)**

NCERT Pg. # 262